



United International University

Department of Computer Science and Engineering

CSE 2213/CSI 219: Discrete Mathematics

Final Examination : Spring 2026

Total Marks: 40 Time: 2 hours

Answer all the 5 questions. Numbers to the right of the questions denote their marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

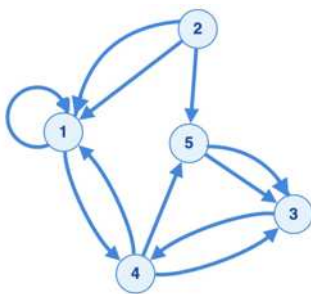
1. (a) An access code can be created in any one of the two formats:

Type I: $LLDS$

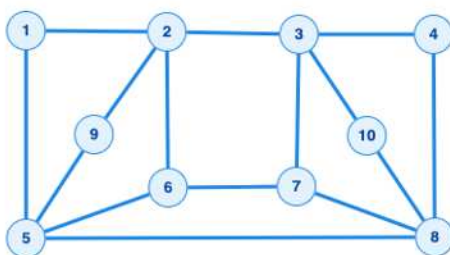
Type II: $LLDS$

where L denotes a capital English letter from A to Z, D denotes a digit from 0 to 9, and S denotes a special symbol from the set $\{ @, \#, \$, \%, \&, *, ? \}$. How many access codes are possible in total? [3]

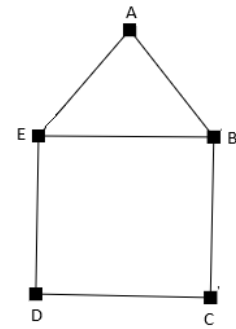
- (b) A university records each student using two categories: birth month and blood group. There are 12 possible birth months and 8 possible blood groups. What is the minimum number of students required to guarantee that at least 4 students have both the same birth month and the same blood group? [2]
- (c) A department has 9 male students and 7 female students. One of the female students is Shakira. How many ways can a 6-member programming team be formed if Shakira must be included in the team? [2]
2. (a) If we represent Facebook friend requests using a graph, where each person is a node and a friendship is an edge, what type of graph does this form? [1]
- (b) Find the in-degrees and out-degrees of each node in the following graph. Then show that the graph follows the Handshaking Theorem. [2+2]



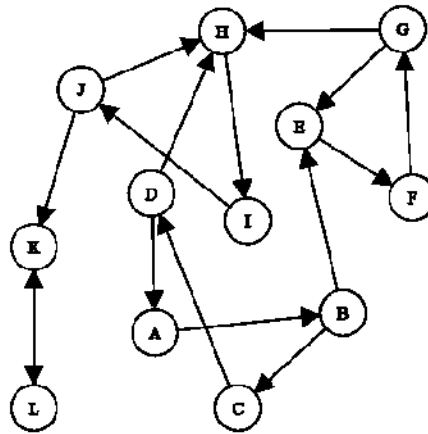
- (c) Determine whether the graph in the following figure is bipartite or not, using the 2-coloring algorithm. [3]



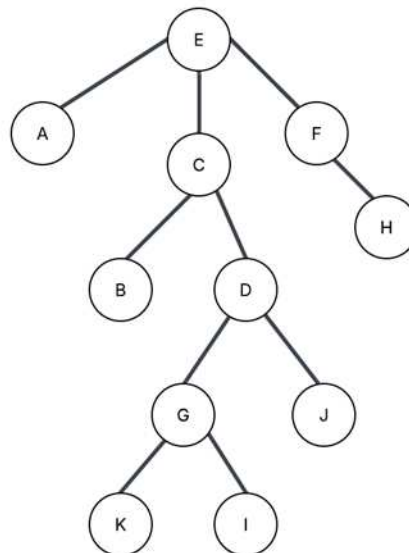
3. (a) Determine whether the following two graphs are isomorphic or not. [4]



- [4]



4. (a) Look at the following tree, and find out the following:



- [2]

France, Portugal, Germany, England, Brazil, Argentina, Spain, Netherlands

(c) Evaluate the following postfix notation:

[2]

$$4\ 3\ 7\ 5\ 1\ +\ -\ \uparrow\ 3\ 4\ 2\ /\ * + -$$

(d) A full m -ary tree has 16 internal vertices and 65 leaves. Find out the value of m , and the number of edges in the tree.

[2]

5. (a) Let the function $f : \mathbb{R} \setminus \{-3\} \rightarrow \mathbb{R} \setminus \{2\}$ be defined by

$$f(x) = \frac{2x+7}{x+3}.$$

Determine whether this function is invertible or not. If it is invertible, find $f^{-1}(7)$.

[3]

(b) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ and $p : \mathbb{R} \setminus \{-2\} \rightarrow \mathbb{R}$ be defined by

$$g(x) = 2x^2 - 5x + 3, \quad p(x) = \frac{3x-1}{x+2}.$$

Find the composite function $(g \circ p)(x)$.

[2]